

Supersede: Diagnosing and Training Memory-Consistent Agents under Fact Supersession

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Abstract

Large language model (LLM) agents increasingly operate over long, multi-session interactions in which facts change: a user moves, a price updates, a plan is revised. Acting correctly requires using the *current* value of a fact and discarding values that have been *superseded*. We isolate this ability on real conversational data and show that it is a distinct, unsolved failure. On the knowledge-update subset of LongMemEval, replacing an agent’s full context with a bounded, self-maintained memory drops accuracy from 92% to 77% even on a frontier model (gpt-5.4), a gap that is statistically significant (paired McNemar $p < 0.005$) and *persists across model scale* while full-context accuracy saturates near 92%. The bottleneck is therefore memory maintenance, not comprehension, and it is not closed by a stronger model. We further show that synthetic, templated supersession is saturated (100%) and thus fails to surface the problem, explaining why it is under-measured. Finally, we release **Supersede**, an open reinforcement-learning environment (built on the verifiers / prime-r1 stack) that turns this measurement into a training signal: agents are rewarded for answering from the current value and penalized for stale ones. The environment reproduces the failure end-to-end and is, to our knowledge, the first trainable environment whose reward targets temporal fact-currency.

Keywords

LLM agents, long-term memory, knowledge updates, reinforcement learning environments, evaluation

1. Introduction

Pre-training has largely consumed the open web, and the frontier of LLM capability has shifted toward agents that act over long horizons using *external memory* rather than a single context window. A defining property of long-horizon interaction is that information is not static: facts are introduced, then later updated or retracted. An assistant that is told a user lives in Detroit, and later that they moved to the suburbs, must answer a subsequent location question with the *current* value. We call the correct handling of such updates *supersession*.

Recent benchmarks have begun to measure long-term memory [1, 2, 3, 4, 5], and a forgetting-aware metric has been proposed to penalize reliance on obsolete information [5]. These works, however, score *frozen* models: they quantify the problem but do not provide a mechanism to *train* it away. In parallel, a line of work applies reinforcement learning to memory agents [6, 7], but rewards final-answer correctness or evidence relevance, never temporal currency. No existing work sits in the intersection: a *trainable* environment whose reward is supersession-correctness.

This paper makes three contributions.

1. **An isolation result.** On real conversational data, we hold memory load fixed and vary only whether the agent has full context or a bounded, self-maintained memory. Bounded memory degrades knowledge-update accuracy significantly (Section 5), isolating memory maintenance — not reading comprehension — as the cause.
2. **A frontier, scale-robust confirmation.** The gap survives on gpt-5.4 (92% $\rightarrow$ 77%, $p = 0.0033$) and does not shrink to zero as the model improves; full-context accuracy, by contrast, saturates. A negative control shows synthetic templated supersession is saturated and therefore uninformative.

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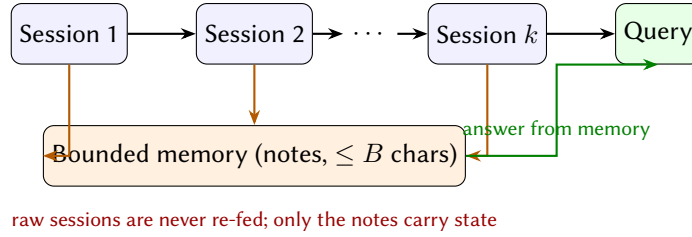


Figure 1: The Supersede rollout. The agent rewrites a bounded notes memory after each session and answers the final query from memory alone. Because history is not re-fed, a superseded value that is not overwritten persists and corrupts the answer.

3. **An open environment.** We release **Supersede**, a verifiers/prime-rl environment with a programmatic supersession-aware reward, validated end-to-end. It reframes the forgetting-aware metric of [5] as a *learning signal*.

2. Related Work

Long-term memory benchmarks. LoCoMo [1] evaluates very long multi-session dialogue but does not annotate a dedicated update category. LongMemEval [2] introduces an explicit knowledge-update question type in which a fact stated in one session is changed in a later one; we adopt this subset as ground truth. MemoryArena [3] shows that models near-perfect on recall benchmarks drop sharply when memory must drive actions, motivating evaluation beyond passive recall. MemoryAgentBench [4] includes a conflict-resolution competency. Memora and its FAMA metric [5] explicitly penalize reliance on superseded or deleted memory. All of these are evaluation-only.

Memory systems and RL for memory. Mem0 [8] manages a memory store with prompted ADD/UPDATE/DELETE operations — a heuristic, not a learned, policy. MemAgent [6] trains a memory agent with RL but rewards only final-answer correctness; LongRLVR [7] adds a verifiable reward on evidence *relevance* within a single long document. Neither rewards *temporal currency* across sessions. Supersede generalizes the verifiable-reward idea of [7] to a time-indexed signal and operationalizes the FAMA metric of [5] as a reward.

Environment infrastructure. We build on the verifiers library and the Prime Intellect Environments Hub [9], which standardize environments for post-training, and are compatible with the emerging OpenEnv standard [10].

3. The Supersede Environment

Task. A task is a multi-session interaction followed by a query about the current value of a fact that changed during the interaction. The agent processes one session at a time and maintains a *bounded* memory (a notes field capped at B characters); crucially, raw sessions are *never re-fed*. After the final session, the agent answers the query using its memory alone. Figure 1 shows the rollout.

Reward. The primary reward is $r_{\text{cur}} = \mathbb{1}[\text{the final answer conveys the current value}]$, scored by a programmatic, ungameable matcher (normalized variant match with a token-overlap fallback), so the environment can be evaluated and trained without an auxiliary judge model. When the superseded values are known (our synthetic tasks, or update-annotated items), a penalty $r_{\text{stale}} = -\mathbb{1}[\text{the answer asserts a superseded value}]$ is added; on gold-only data it is inert. The combined reward is $r = r_{\text{cur}} + \lambda r_{\text{stale}}$. This makes the forgetting-aware accuracy of [5] a dense training signal rather than a post-hoc score.

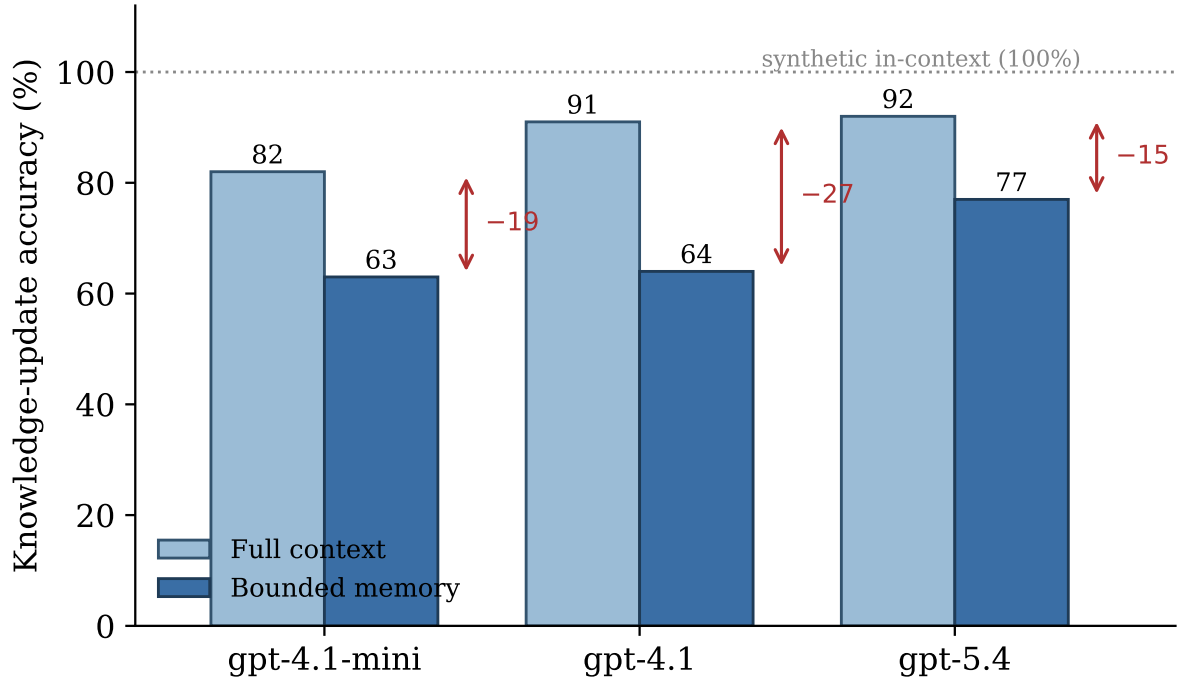


Figure 2: Knowledge-update accuracy: full context vs. bounded memory, across three models. Stronger models read context better (full-context rises to 92%) but bounded-memory accuracy does not catch up; the gap is the cost of memory maintenance under updates. The dotted line marks synthetic in-context performance (100%), which is saturated and uninformative.

Data. Tasks are drawn from the LongMemEval knowledge-update subset (real conversational supersession) and from a procedural generator of synthetic timelines used for controls and ablations. The environment is implemented as a `verifiers MultiTurnEnv`; the prompt assembly enforces the no-re-feed property, and rollouts terminate when the answer is produced.

4. Experimental Setup

Data and grading. We use all 78 knowledge-update questions from the LongMemEval *oracle* split (evidence sessions only). Unless noted, answers are graded by an LLM judge following the LongMemEval protocol; the environment’s intrinsic reward uses the programmatic matcher.

Conditions. `FULL-CONTEXT` places every evidence session in the model context and asks the question (an upper bound where memory is not the bottleneck). `BOUNDED-MEMORY` is the Supersede agent of Section 3 with $B = 300$ characters.

Models. We evaluate gpt-4.1-mini, gpt-4.1, and the frontier gpt-5.4, at temperature 0.

Significance. Because both conditions are run on the same questions, we use a paired McNemar test.

5. Results

Bounded memory degrades supersession, significantly. Table 1 and Figure 2 report the main result. For gpt-4.1-mini, accuracy falls from 82% to 63%; the paired test counts 19 questions that full-context answers correctly but bounded-memory does not, against only 4 in the reverse direction ($p = 0.0035$). Roughly 13% of questions are wrong even in full context, confirming that real updates are genuinely hard.

Table 1

LongMemEval knowledge-update ($n = 78$). The full-context $\rightarrow$ bounded-memory gap is significant on both the small and the frontier model.

Model	Full context	Bounded memory	Paired McNemar
gpt-4.1-mini	82%	63%	$p = 0.0035$
gpt-4.1	91%	64%	—
gpt-5.4	92%	77%	$p = 0.0033$

The gap persists on the frontier and does not close with scale. gpt-5.4 raises full-context accuracy to 92% but bounded-memory accuracy only to 77% ($p = 0.0033$; 13 vs. 1). Bounded-memory accuracy across the family (63%, 64%, 77%) improves far more slowly than full-context accuracy (82%, 91%, 92%): scaling the model helps memory maintenance only partially, while comprehension saturates. The failure is therefore a property of the memory mechanism, not of model capability.

Failure modes. Bounded-memory errors are dominated by the relevant fact being compressed away or not overwritten. Asked “How many Korean restaurants have I tried?” (gold: *four*), the agent answers “you haven’t mentioned any.” Asked “Where did Rachel move to?” (gold: *the suburbs*), gpt-5.4 answers “no information about Rachel.” In both, the model is competent but its memory no longer contains the updated fact.

Synthetic supersession is saturated. As a negative control, we ran the same models on procedurally generated, templated supersession timelines (explicit “ X now Y ” updates) with full history in context: accuracy is 100% across configurations. Frontier models trivially resolve clean, explicit updates by a last-mention scan; the failure only appears with bounded memory on natural, paraphrased data. This explains why supersession is under-measured and motivates the use of real data.

Environment validation. Running the released environment end-to-end on all 78 questions, every rollout terminates cleanly and the intrinsic programmatic reward yields 57.7% for gpt-4.1-mini, consistent with the 63% measured by the LLM judge (the matcher is stricter). The environment thus faithfully reproduces the failure it is designed to train against.

6. Discussion

The result reframes supersession from a comprehension problem to a memory-policy problem. Stronger models read updated context well, but when they must *decide what to keep* under a bounded budget, they discard or fail to overwrite the value that later matters. This is precisely the regime that production memory systems operate in, and precisely where a heuristic UPDATE/DELETE policy [8] is brittle. Because the gap does not close with scale, waiting for larger models is not a solution; a memory policy that is *trained* to preserve currency is. Supersede provides the environment and the verifiable reward to do so.

7. Limitations

We report a benchmark-and-environment result, not a trained-policy result: demonstrating that the reward improves a policy requires RL training and is left to future work. Grading by LLM judge introduces some noise (a few bounded-memory failures are judge strictness, e.g. “over 25” vs. gold “25”); the paired significance survives a handful of such flips. We use the oracle split; the larger distractor splits should widen the gap and are a natural next measurement. Finally, the synthetic generator is a control, not a training target.

8. Conclusion

Agents fail at supersession not because they cannot read updates but because they cannot maintain them under a bounded memory — a gap that is significant, reproducible on a trusted benchmark, and unclosed by frontier scale. We release Supersede, an open environment that turns this forgetting-aware measurement into a training signal, as a step toward agents whose memory stays current.

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Declaration on Generative AI

During the preparation of this work, the author used a code-assistant (Claude) for drafting prose, generating figures, and implementing and debugging the environment and evaluation code. All experiments, numbers, and claims were executed and verified by the author, who reviewed and edited the content and takes full responsibility for the publication’s content.

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